higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# NATIONAL CERTIFICATE (VOCATIONAL)

# ELECTRONIC CONTROL AND DIGITAL ELECTRONICS
## NQF LEVEL 2

(12041022)

**13 November 2019 (Y-Paper)**
**13:00–16:00**

**Calculators may be used.**

**This question paper consists of 6 pages.**

Copyright reserved

**TIME: 3 HOURS**
**MARKS: 100**

---

**INSTRUCTIONS AND INFORMATION**

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Write neatly and legibly.

---

Copyright reserved 

## QUESTION 1

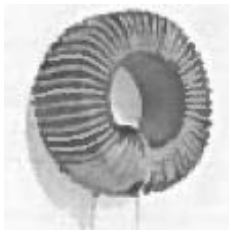
1.1 Explain how the number of electrons is determined in an atom's shell. (4)

1.2 Covalent bonding is the mutual sharing of electrons between atoms to complete the valence shell.

Use a diagram to illustrate the concept of covalent bonding exhibited by water. (5)

1.3 Name each of the following components:

1.3.1 

1.3.2 

(2 × 1) (2)

1.4 Briefly describe the formation of N-type material. (4)

1.5 Explain reverse biasing of the PN junction device. (2)

1.6 Name ONE application of a zener diode. (1)

1.7 Make a neat, labelled circuit diagram of a half wave rectifier that uses a single diode. (4)

1.8 Name THREE regions in which a transistor can operate. (3)

**[25]**

## QUESTION 2

2.1 What is an *operational amplifier?* (2)

2.2 Explain the meaning of majority and minority charge carriers. (4)

2.3 State the use of each of the following tools:

2.3.1 Wire stripper

2.3.2 Soldering iron

2.3.3 Side cutter

2.3.4 Long nose pliers

(4 × 1) (4)

2.4 Explain THREE dangers of electrostatic charge. (3)

2.5 List FIVE safety precautions to be taken when handling an electrostatic-sensitive device. (5)

2.6 TWO capacitors of 10 µF and 4,7 µF are connected in parallel.

Calculate the total capacitance. (2)

**[20]**

## QUESTION 3: DIGITAL ELECTRONICS

3.1 Use sketches to differentiate between an analogue and a digital signal. (4)

3.2 Digital circuits represent two bit values, zero and ONE, using two voltage levels.

Briefly explain how voltage levels are used to represent bits. (4)

3.3 Define each of the following digital terms:

3.3.1 Timing diagram

3.3.2 Clock signal

3.3.3 Positive logic

(3 × 1) (3)

3.4 Convert the decimal number $42_{10}$ to a binary number. (4)

Copyright reserved 

3.5 Name FOUR of the seven categories of basic logic gates. (4)

3.6 Study FIGURE 1 below and answer the questions.

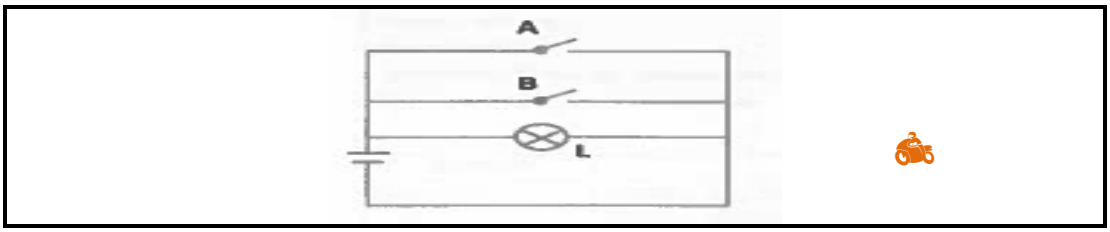



**FIGURE 1**

3.6.1 Name the logic gate shown in FIGURE 1. (2)

3.6.2 Describe the operation of the circuit in FIGURE 1 (2)

3.6.3 Use a truth table to substantiate your answer in 3.6.2 above. (4)

3.6.4 Explain what will happen to the bulb if switch A is faulty and behaves like an open circuit (4)

3.6.5 Use a truth table to substantiate your answer in 3.6.4 above (4)

**[35]**

## QUESTION 4: BASIC COMPUTER COMPONENTS

4.1 Choose a description from COLUMN B that matches a word in COLUMN A. Write only the letter (A–D) next to the question number (4.1.1–4.1.3) in the ANSWER BOOK.

| COLUMN A | | COLUMN B | |
|---|---|---|---|
| 4.1.1 | Hardware | A | consists of the programs that enable a computer to perform a specific task |
| 4.1.2 | Firmware | B | is software used for the limited purpose of changing the overall behavior of hardware or other software |
| 4.1.3 | Software | C | physical part of a computer which consists of all digital circuitry |
| | | D | software that is embedded in a hardware device |
| | | E | used for a specific purpose, such as word processing, spreadsheet, database or accounting programs |

(3 × 1) (3)

4.2 Explain the function of each of the following computer ports:

4.2.1 VGA (video graphic array)

4.2.2 RJ-45

(2 × 1) (2)

4.3 Describe how data are stored in a flash memory cell. (5)
**[10]**

**QUESTION 5: TRANSDUCERS USED IN PROCESS CONTROL**

5.1 Name FOUR physical conditions that can be measured by non-electrical instruments. (4)

5.2 Draw a construction of an NPN phototransistor. (4)

5.3 Name TWO limitations of non-electrical measuring instruments. (2)
**[10]**

**TOTAL: 100**

Copyright reserved